

1 Moore-Penrose Pseudoinverse

Recall that the pseudoinverse $\mathbf{X}^+ = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ appears as part of the analytic solution of linear regression. In this problem, we will explore some of the properties of this pseudoinverse.

Problem 1. Let $\mathbf{X} \in \mathbb{R}^{N \times M}$ and assume $\mathbf{X}^\top \mathbf{X}$ is nonsingular.

1. Show that $\mathbf{X}^+ \mathbf{X} = \mathbf{I}$.
2. What is the dimension of $\mathbf{I}$?

Problem 2. The pseudoinverse $\mathbf{X}^+ = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is also known as a left-inverse.

1. Why do we call $\mathbf{X}^+$ a left-inverse?
2. What is a natural way to define a right-inverse?
3. What do you need to assume with that definition?

Problem 3. Let $\mathbf{X} \in \mathbb{R}^{N \times N}$ and assume both $\mathbf{X}^\top \mathbf{X}$ and $\mathbf{X}$ are nonsingular.

1. By considering $\mathbf{X} \mathbf{X}^+ \mathbf{X} \mathbf{X}^{-1}$ or otherwise, show that $\mathbf{X} \mathbf{X}^+ = \mathbf{I}$.
2. By considering $\mathbf{X}^{-1} \mathbf{X} \mathbf{X}^+$ or otherwise, show that $\mathbf{X}^+ = \mathbf{X}^{-1}$.

2 Vector Calculus

Many modern machine learning techniques rely on using gradient information to search for a model. Typically, this involves differentiating a scalar function with respect to a vector. Take linear regression as an example: we differentiate the error function E with respect to the parameters $\mathbf{w}$. If $\mathbf{w} \in \mathbb{R}^N$, then this is formally written as

$$\nabla_{\mathbf{w}} E(\mathbf{w}) = \left[\frac{\partial E}{\partial w_1} \quad \cdots \quad \frac{\partial E}{\partial w_N} \right]^\top.$$

One method to derive $\nabla_{\mathbf{w}} E(\mathbf{w})$ is to compute the partial derivatives individually and combine it back into a vector. For example, given a function $f(\mathbf{w}) = \sum_i a w_i$ for some constant $a \in \mathbb{R}$, then the i -th partial derivative $\partial f / \partial w_i = a$. Hence

$$\nabla_{\mathbf{w}} f(\mathbf{w}) = [a \quad \cdots \quad a]^\top.$$

In this problem, we shall derive some of the basic derivative rules in vector calculus. Although in practice, most people simply refer to the Matrix Cookbook for these rules, it is essential that you know how these rules are derived.

Problem 1. Let $\mathbf{w} \in \mathbb{R}^N$ and $\mathbf{x} \in \mathbb{R}^N$. Consider $f(\mathbf{w}) = \mathbf{w}^\top \mathbf{x}$.

1. For each $i \in [N]$, show that the partial derivative $\partial f / \partial w_i = x_i$.
2. Show that $\nabla_{\mathbf{w}} f(\mathbf{w}) = \mathbf{x}$.

Problem 2. Let $\mathbf{w} \in \mathbb{R}^N$. Consider $f(\mathbf{w}) = \mathbf{w}^\top \mathbf{w}$.

1. For each $i \in [N]$, find the partial derivative $\partial f / \partial w_i$.
2. Show that $\nabla_{\mathbf{w}} f(\mathbf{w}) = 2\mathbf{w}$.

Problem 3. Let $\mathbf{w} \in \mathbb{R}^N$ and $g(\mathbf{w}), h(\mathbf{w})$ be scalar functions of $\mathbf{w}$. Consider $f(\mathbf{w}) = g(\mathbf{w}) + h(\mathbf{w})$.

1. For each $i \in [N]$, find the partial derivative $\partial f / \partial w_i$.
2. Show that $\nabla_{\mathbf{w}} f(\mathbf{w}) = \nabla_{\mathbf{w}} g(\mathbf{w}) + \nabla_{\mathbf{w}} h(\mathbf{w})$.

Problem 4. Let $\mathbf{w} \in \mathbb{R}^N$ and $\mathbf{A} \in \mathbb{R}^{N \times N}$ such that $\mathbf{A}^\top = \mathbf{A}$. Consider the function $f(\mathbf{w}) = \mathbf{w}^\top \mathbf{A} \mathbf{w}$.

1. For each $i \in [N]$, find the partial derivative $\partial f / \partial w_i$.
2. Show that $\nabla_{\mathbf{w}} f(\mathbf{w}) = 2\mathbf{A} \mathbf{w}$.

Problem 5. Recall that the error function for the L2-regularized linear regression is

$$E(\mathbf{w}) = \frac{1}{2} (\mathbf{t} - \mathbf{X} \mathbf{w})^\top (\mathbf{t} - \mathbf{X} \mathbf{w}) + \frac{\lambda}{2} \|\mathbf{w}\|_2^2.$$

Find $\nabla_{\mathbf{w}} E(\mathbf{w})$.

3 Probabilistic Inference

A crucial goal of machine learning is to make inferences about a process based on the data collected from the same process. When deriving the error function of linear regression in lecture, this process is assumed to be the distribution $\mathcal{N}(t|\mathbf{w}^\top \mathbf{x}, \alpha^{-1})$ and the data is assumed to be drawn from this process in an i.i.d. fashion. Since the inference is made using probabilistic arguments, we call this a probabilistic inference.

In this problem, we shall find an optimal strategy for guessing the biasness of a coin. Suppose we are given a coin with unknown bias q . For each flip, we obtain heads with probability q and tails with probability $1 - q$. Our goal is to find a best guess for q , which we will denote as $\hat{q}$.

Problem 1. We flip the coin 10 times and observe $\mathcal{D} = \{H, T, T, H, T, H, H, H, T, H\}$. Intuitively, we guess that $\hat{q} = 0.6$ because 6 out of the 10 flips are heads. Justify our guess with probabilistic reasoning using the following guiding questions.

- Find the log-likelihood of observing the sequence of flips given that the true coin has a fixed bias q .
- Show that the $\hat{q}$ that maximizes the likelihood is 0.6.

Problem 2. Our coin provider later tell us that q can only be either 0.3, 0.5 or 0.7. In other words, the space of all possible process shrinks from $\mathcal{H} = [0, 1]$ to $\mathcal{H} = \{0.3, 0.5, 0.7\}$. Find the maximum likelihood estimator(s) $\hat{q}$.

Problem 3. The coin provider then show us five coins, one of which has $q = 0.3$, one of which has $q = 0.5$, and three of which has $q = 0.7$. After drawing exactly one of these five coins uniformly at random, you flip it 10 times and obtained the sequence $\mathcal{D} = \{H, T, T, H, T, H, H, H, T, H\}$. Find the maximum a posteriori estimator(s) $\hat{q}$.